



ABSTRACT

NFC ACCESS CONTROL

IDLE Scan card
AUTH 30 s window
RUNNING Current detected

NON-INTRUSIVE SENSING

```
// True RMS over 2000 samples
rmsV = sqrt((sum(samples.^2) / N)
A = (rmsV / 330) * 2000
```

LIVE WEB DASHBOARD

```
GET /api/status
GET /api/power
POST /api/set_threshold
```

AMBIENT CANCELLATION

```
rawAmps - gAmbient = netAmps
// EMA filter  $\alpha = 0.15$ 
out =  $\alpha \cdot \text{net} + (1-\alpha) \cdot \text{prev}$ 
```

TELEGRAM ALERTS

OFFLINE EXCEL EXPORT

- Sheet 1 · Power Log (raw + net A, W)
- Sheet 2 · Runtime by Day
- Sheet 3 · Operator Summary
- Sheet 4 · Event Log

EVENT-DRIVEN STATE MACHINE

